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Lifetime Measurements for KrF Excimer Using Photoassociative Pumping

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Various kinetic parameters of the KrF excimer generated by the use of photoassociative pumping have been measured under simple and controllable conditions. The population transfer rate coefficient between the $B^2\Sigma^+$ and $C^2\Pi$ states has been directly determined to be $(1.2\pm0.3)\times10^{-10}$ cm^3/s for Kr buffer gas. The radiative lifetime of the $C^2\Pi$ state and the quenching rate coefficients for F_2 and Kr gases are also given as 60 ± 10 ns, $(2.5\pm1)\times10^{-10}$ cm^3/s, and $(7\pm3)\times10^{-12}$ cm^3/s, respectively.

KEYWORDS: KrF, photoassociation, collision-induced absorption, excimer laser, kinetics

§1. Introduction

In modeling the kinetic processes in the medium of rare gas halide excimer lasers,[1,2] it is a common knowledge that the population transfer process between the low vibrational $B^2\Sigma^+$ and $C^2\Pi$ states, the so-called B/C mixing process, is a key process which strongly influences the quantitative estimation of the characteristics for the laser medium, especially of the extraction efficiency of the lasing photons. Such an estimation is especially important for the KrF excimer laser in order to establish its physical basis as a high power and high efficiency UV radiation source scalable to a laser fusion driver.

So far, there have been several reports on the investigations of the B/C mixing process for KrF. Pummer *et al.*[3] obtained a B/C mixing time, τ_{BC}, of $\tau_{BC}\leqq1$ ns at 2 atm He bufgfer gas pressure by observing the sidelight fluorescence from a discharge KrF laser oscillator. By measuring the gain in a discharge medium for a picosecond pulse, Szatmari and Shafer[4] estimated the B/C mixing time to be shorter than 10 ps at 2.3 atm He buffer gas. Kannari *et al.*[5] estimated the B/C mixing rate coefficient for Ar buffer gas as 5×10^{-10} cm^3/s by fitting their simulation to experimental characteristics of sidelight fluorescence from an electron-beam-pumped amplifier.

However, most of those previous experiments were carried out using the laser device itself. In such experiments, many excitation and quanching processes occur at the same time, influencing each other in a very complex manner, and it is extremely difficult to determine one parameter independently from others. In order to obtain a reliable value of the B/C mixing rate, the experimental measurements should be carried out systematically under controllable conditions.

Although systematic studies have been performed for XeF and XeCl excimers[6–8] using optical pumping techniques, there has been no study done on the KrF excimer, probably because such state selective pumping was believed to be very difficult for the KrF excimer because of its perfectly repulsive potential curve in the ground state. Recently, however, we have succeeded in selectively pumping the KrF excimer in the lasing state by photoassociation at 248 nm.[9] This success has made it possible for us to determine directly the B/C mixing rate for the KrF excimer by observing the fluorescence lifetime and its pressure dependence. In the present paper, we report on the lifetime measurements using this photoassociative pumping, which directly yields the B/C mixing rate coefficient as well as the radiative lifetime and the quenching rate coefficients for F_2 and Kr gases for the C state.

§2. Experiment

The gas mix of Kr and F_2 was irradiated by 248 nm radiation generated by a discharge-pumped KrF laser with an unstable resonator arrangement. Uniform irradiation was obtained in a sample cell by collimating the laser beam and passing it through an iris of 3 mm diameter. The irradiated intensity was kept at 10 MW/cm^2 throughout the measurements. The fluoresence was detected by a photomultiplier (Hamamatsu H3177) after passing through a monochromator, and the signal was averaged with a transient digitizer (Tektronix 7912AD) interfaced to a microcomputer. The time resolution of the present system is 10 ns, which is limited by the laser pulse width.

The temporal behaviors of the fluorescence were observed for the $B^2\Sigma^+$–$X^2\Sigma^+$ and $C^2\Pi$–$A^2\Pi$ transitions by selecting the monitoring wavelength at 245 and 280 nm, respectively. The B–X transition was monitored at the wavelength 3 nm shorter than the band center wavelength in order to reduce the influence of strong excitation radiation.

§3. Results

Figure 1 shows the temporal profiles of the B–X and C–A fluorescence. The partial pressures of Kr and F_2 gases are 9 and 1 Torr, respectively. It is clearly seen that the B–X fluorescence shows a peculiar double exponential feature in contrast to the C–A fluorescence showing a single exponential decay. The B–X fluorescence decays rapidly during the first 30 ns from the peak, due to the decrease in the B state population through radiative

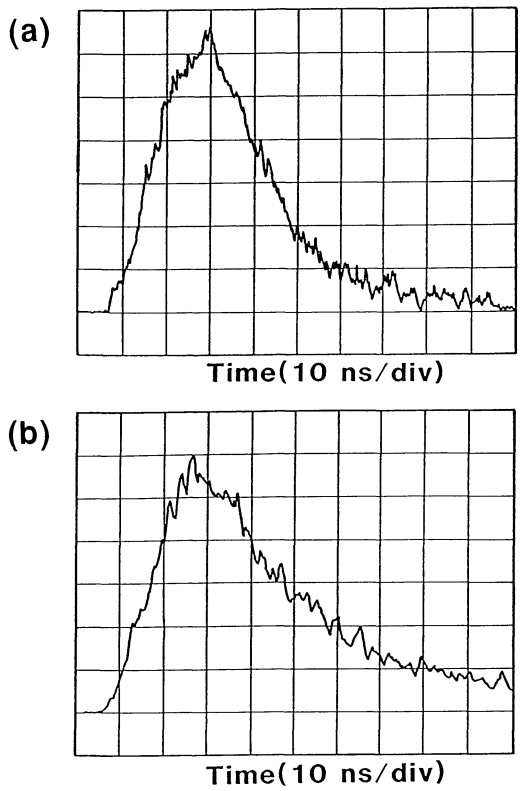


Fig. 1. Decay curves of B–X (a) and C–A (b) transitions of KrF excimer observed by photoassociation in Kr/F_2 gas mix. Partial pressure of Kr and F_2 are 9 and 1 Torr, respectively.

decay, collisional quenching, and B/C mixing. The tail portion of the B–X fluorescence curve shows slow decay with a time constant of 25 ns, which is coincident with the decay time of the C–A fluorescence. This behavior reflects the fact that the populations of the B and C states, which lie close together in energy, are rapidly transferred from one state to the other by collisions with the buffer gas.

We can describe the situation much more clearly by solving the rate equations for the populations of the B and C states. Under the present simple pumping condition, the analytical solutions of the rate equations are given[6,10] as below:

$$N^{(B)}=D^{(B)}\exp(-\lambda_1 t)+E^{(B)}\exp(-\lambda_2 t), \quad (1)$$

and

$$N^{(C)}=D^{(C)}\exp(-\lambda_1 t)+E^{(C)}\exp(-\lambda_2 t), \quad (2)$$

where

$$\lambda_{1,2}=(\beta+\gamma)/2\pm[\{(\beta-\gamma)/2\}^2+k_{BC}k_{CB}n_{Kr}^2]^{1/2}, \quad (3)$$

$$\beta=A_B+k_{F_2}^{(B)}n_{F_2}+k_{Kr}^{(B)}n_{Kr}+k_{2Kr}^{(B)}n_{Kr}^2+k_{BC}n_{Kr}, \quad (4)$$

and

$$\gamma=A_C+k_{F_2}^{(C)}n_{F_2}+k_{Kr}^{(C)}n_{Kr}+k_{2Kr}^{(C)}n_{Kr}^2+k_{CB}n_{Kr}. \quad (5)$$

Here, k_{BC} is the B/C mixing rate coefficient of the B state to the C state, k_{CB} that for vice versa, and β and γ denote the decay rates of the B and C states, respectively, including radiative, quenching, and mixing processes. A k_{F_2}, k_{Kr}, and k_{2Kr} are the radiative decay rate and rate coefficients for F_2 quenching, two body Kr quenching, and three body Kr quenching for the suffixed state, respectively. Under the partially mixed condition, the B state population $N^{(B)}$ mainly decays through the fast λ_1 term, and the C state population $N^{(C)}$ mainly through the slow λ_2 term. Then, the double exponential feature should be relatively easy to distinguish in the B–X fluorescence, but should be difficult in the C–A fluorescence. These behaviors well explain the observed results shown in Fig. 1. By measuring the pressure dependence of the fast or slow decay rate, we can obtain the rate coefficients for B/C mixing and also for other kinetic processes. In this study, we have investigated the slow decay rate by observing the C–A fluorescence, for which our time resolution is sufficiently high.

Firstly, we show the pressure dependence for the F_2 gas. The Stern-Volmer plot of the C–A fluorescence signal is shown in Fig. 2. The pressure of Kr gas is held at 9 Torr. The $1/\tau$ increases linearly up to $8\times10^7\ s^{-1}$ with change of the partial pressure of F_2 gas from 1 to 5 Torr. This linear behavior is explained well by a model that the

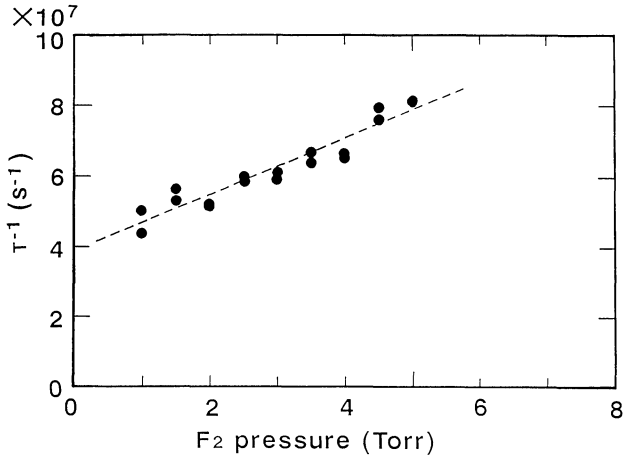


Fig. 2. Stern-Volmer plot for the decay rate of the C state. The pressure of F_2 gas was varied from 1 to 5 Torr. The Kr partial pressure was held at 9 Torr.

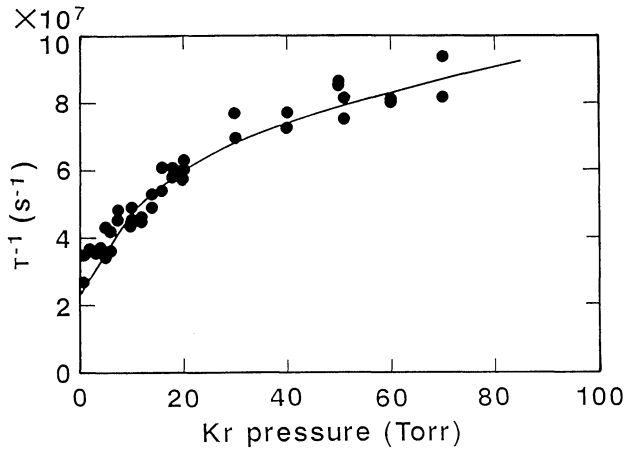


Fig. 3. The experimental plot of the Kr pressure dependence for the decay rate of the C–A fluorescence. The F_2 partial pressure was held at 1 Torr.

collisional process of KrF with F_2 gas is a simple quenching process. By fitting the results, we obtained an F_2 quenching rate coefficient of $(2.5\pm1)\times10^{-10}$ cm^3/s for the C state.

Secondly, the pressure dependence for the Kr gas is shown in Fig. 3. The black circles represent the experimental results. The F_2 partial pressure was kept at 1 Torr. The $1/\tau$ increases up to 8×10^7 s^{-1} by varying the Kr pressure from 1 to 70 Torr. Its behavior is completely different from that for F_2 quenching: τ^{-1} increases linearly in the low-pressure region up to 10 Torr, and shows a saturating behavior in the higher-pressure region. This behavior cannot be explained by a simple quenching mechanism, but is described well by the model in which B/C mixing becomes faster with increase of the buffer gas pressure. Actually, as seen in eqs. (3)–(5), if the quenching processes due to Kr gas are negligible in the decay process, the model predicts that the slow decay rate λ_2 tends to a mean value of decay rates of the B and the C state at zero Kr pressure.

The Kr pressure dependence was analyzed by using eqs. (3)–(5). By fitting the observed curve to λ_2, we obtained the radiative lifetime of the C state, the two body quenching rate coefficient for Kr, and the B/C mixing rate coefficient for Kr buffer gas. The radiative lifetime of the B state and three-body quenching rate coefficient were fixed to 6.8 ns[11,12] and 2.9×10^{-31} cm^6/s,[5,13] respectively. We assumed the same quenching rate coefficients for both B and C states. To separate k_{BC} and k_{CB}, we used the detailed balance relation,

$$k_{CB}=\exp(-\Delta E/kT)\cdot k_{BC}. \quad (6)$$

Here, ΔE is an energy difference between the B and C state. Since it is known that the C state lies 80 cm^{-1} lower than the B state,[14] the ratio k_{CB}/k_{BC} becomes 0.7 at room temperature. The fitted curve is superimposed on the observed results in Fig. 3 by a solid line, which is in good agreement with the observed results.

§4. Discussion

The parameters thus obtained are summarized in Table I along with the previously known values. Here, the B/C mixing time reported in refs. 3 and 4 are converted to the corresponding rate coefficient. It must be mentioned that, in the present work, the parameters have been obtained quantitatively under simple and controllable conditions.

Although the B/C mixing rate coefficients cited in Table I are obtained for different gases, it would be meaningful to compare them with our value. We should especially compare the value given by Szatmari and Shafer,[4] which is too large compared to our value. They presented the value to explain a large fluorescence depression induced by a picosecond laser using a commercial discharge device. However, in such a high-pressure discharge medium, many effects from various kinetic processes are overlapped on the behavior of the fluorescence, and it would be very difficult to obtain one kinetic parameter independently from others only from the fluorescence depression. We believe that the observed large depression[4] is explained not simply by an extremely fast B/C mixing process but by the combination of reasonably fast B/C mixing and other processes.

Table I. Obtained kinetic parameters for the KrF excimer. Buffer Gases for B/C mixing are denoted in parentheses.

	B/C mixing $\times10^{-10}$ cm^3/s	Radiative lifetime in ns	F_2 quenching $\times10^{-10}$ cm^3/s	Kr quenching $\times10^{-12}$ cm^3/s
This study	1.2±0.3 (Kr)	60±10	2.5±1	7±3
Other	>0.2 (He)[a]	75[d]	7.8[e]	20[h]
studies	>17 (He)[b]		5.7[f]	2[c]
		5 (Ar)[c]		4.8[g]
			3.4[c]	

a) Ref. 3. b) Ref. 4. c) Ref. 5. d) Ref. 11. e) Ref. 15. f) Ref. 13. g) Ref. 12. h) Ref. 16.

By using our B/C mixing rate, we can estimate the B/C mixing time constants in a real KrF laser amplifier. Such an estimation is especially important to assess the KrF laser medium of atmospheric pressure with high Kr concentration,[17,18] which is proposed as a promising candidate for a large scale amplifier medium of an electron-beam-pumped KrF laser system for a laser fusion driver.[19] In the Kr buffer gas of atmospheric pressure, our present B/C mixing rate coefficient demonstrates B→C and C→B mixing time constants of 300 and 400 ps, respectively. This means that the pulse width of the incident laser beam must be longer than at least 400 ps in the amplifier with an atmospheric high Kr concentration medium in order to efficiently extract the energy from both B and C states.

Among other parameters listed in Table I, we would like to note that the radiative lifetime of the C state has been obtained experimentally for the first time. The obtained lifetime is consistent with the theoretical value obtained by Dunning and Hay.[11] For the F_2 quenching coefficient, our value seems to be small compared with other values, but is in agreement with that of Kannari *et al.*[5] within the error limit. This correspondence would be reasonable considering the fact that Kannari *et al.* obtained the parameter by modifying the results of Rokni *et al.*[15] using the B/C mixing model.

In conclusion, we have investigated the B/C mixing process of KrF excimer using photoassociative pumping. The radiative lifetime and quenching rate coefficients for the C state have also been measured. This study has demonstrated the usefulness of the photoassociation technique for the kinetic study of the excimer laser medium. Finally, we should mention that is should be possible to extend the use of photoassociative pumping toward the investigation of the vibrational relaxation of the KrF excimer by selectively pumping the various vibrational states with a powerful tunable UV laser source.

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